

Network Troubleshooting Lab 003

DHCP Troubleshooting

Scenario:

A user reports difficulty connecting to network resources. This troubleshooting lab focuses on analyzing DHCP behavior and verifying that a device receives valid network configuration information automatically.

Document Information

Category: Networking / IT Support

Lab Type: Troubleshooting Case

Environment: MacBook Pro - macOS

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1. Troubleshooting Methodology

Followed a structured troubleshooting approach:

1. Identify the problem
2. Establish a theory of probable cause
3. Test the theory
4. Analyze results
5. Document findings

2. Step 001 — Review Network Configuration

Command Used

networksetup -getinfo Wi-Fi

Purpose

Used to review the current network configuration and verify how the device receives its IP address information.

Information reviewed:

- IP Address
- Subnet Mask
- Router
- DHCP configuration

Evidence

```
Last login: Mon Oct 9 20:29:24 on ttys000
ridleyherisse@Ridleys-MacBook-Pro ~ % networksetup -getinfo Wi-Fi

DHCP Configuration
IP address: 10.0.0.207
Subnet mask: 255.255.255.0
Router: 10.0.0.1
Client ID:
IPv6: Automatic
IPv6 IP address: none
IPv6 Router: none
Wi-Fi ID: 3c:0e:30:33:b6:6d
ridleyherisse@Ridleys-MacBook-Pro ~ %
```

Findings

The network configuration was reviewed to determine whether the device successfully obtained valid network settings from DHCP.

3. Step 002 — Verify IP Address Assignment

Command Used

Ifconfig en0

Purpose

Used to examine the active network interface and verify the assigned IP address.

Evidence

```
status: inactive
utun0: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
    inet6 fe80::af27:39e9:de17:1645%utun0 prefixlen 64 scopeid 0xf
    ndo options=281<PERFORMNUD,DAD>
utun1: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 2800
    inet6 fe80::5616:4fc:47eb:1f6c%utun1 prefixlen 64 scopeid 0x10
    ndo options=281<PERFORMNUD,DAD>
utun2: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1000
    inet6 fe80::ce81:b1c:bd2c:49%utun2 prefixlen 64 scopeid 0x11
    ndo options=281<PERFORMNUD,DAD>
utun3: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
    inet6 fe80::f377:9484:a35a:ca86%utun3 prefixlen 64 scopeid 0x12
    ndo options=281<PERFORMNUD,DAD>
utun4: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
    inet6 fe80::c682:e93b:7892:55c3%utun4 prefixlen 64 scopeid 0x13
    ndo options=281<PERFORMNUD,DAD>
utun5: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
    inet6 fe80::8893:5a51:cfcc:f8c9%utun5 prefixlen 64 scopeid 0x14
    ndo options=281<PERFORMNUD,DAD>
utun6: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
    inet6 fe80::af27:8863:c81:acbd%utun6 prefixlen 64 scopeid 0x15
    ndo options=281<PERFORMNUD,DAD>
ridleyherisse@Ridleys-MacBook-Pro ~ % networksetup -listallhardwareports

Hardware Port: Ethernet Adapter (en3)
Device: en3
Ethernet Address: 32:48:3e:2b:49:60

Hardware Port: Ethernet Adapter (en4)
Device: en4
Ethernet Address: 32:48:3e:2b:49:61

Hardware Port: Thunderbolt Bridge
Device: bridge0
Ethernet Address: 36:32:a2:89:68:40

Hardware Port: Wi-Fi
Device: en0
Ethernet Address: 3c:86:38:33:b6:6d

Hardware Port: Thunderbolt 1
Device: en1
Ethernet Address: 36:32:a2:89:68:40

Hardware Port: Thunderbolt 2
Device: en2
Ethernet Address: 36:32:a2:89:68:44

VLAN Configurations
=====
ridleyherisse@Ridleys-MacBook-Pro ~ % ifconfig en0
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=4463<RXCSUM,TXCUM,TSO4,TSO6,CHANNEL_ID,PARTIAL_CSUM,ZEROINVERT_CSUM>
    ether 3c:86:38:33:b6:6d
    inet6 fe80::1cad:84e2:16c0:d323%en0 prefixlen 64 secured scopeid 0xc
    inet 10.0.0.207 netmask 0xfffff000 broadcast 10.0.0.255
    inet6 2681:195:c608:f9c0:1c24:dd69:2f50:8758 prefixlen 64 autoconf secured
    inet6 2681:195:c608:f9c0:860d:8aa2:a82:f88b prefixlen 64 autoconf temporary
    inet6 2681:195:c608:ffc0:c49b prefixlen 64 dynamic
    ndo options=281<PERFORMNUD,DAD>
    media: autoselect
    status: active
ridleyherisse@Ridleys-MacBook-Pro ~ %
```

Findings

The active Wi-Fi interface (en0) was assigned a valid IPv4 address (Censored), subnet mask, and broadcast address. This confirmed that the device successfully received network configuration information and was connected to the local network.

4. Step 003 — Renew DHCP Lease

Command Used

```
sudo ipconfig set en0 DHCP
```

Purpose

Used to request updated network configuration information from the DHCP server and renew the device's network lease.

Evidence

```
utun6: flags=8851<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
    inet6 fe80::af27:8863:c81:acbb%utun6 prefixlen 64 scopeid 0x18
    nd6 options=201<PERFORMNUD,DAD>
ridleylherisse@Ridleys-MacBook-Pro ~ % networksetup -listallhardwareports

Hardware Port: Ethernet Adapter (en3)
Device: en3
Ethernet Address: 32:48:3e:2b:49:60

Hardware Port: Ethernet Adapter (en4)
Device: en4
Ethernet Address: 32:48:3e:2b:49:61

Hardware Port: Thunderbolt Bridge
Device: bridge0
Ethernet Address: 36:32:a2:89:68:40

Hardware Port: Wi-Fi
Device: en0
Ethernet Address: 3c:06:30:33:b6:6d

Hardware Port: Thunderbolt 1
Device: en1
Ethernet Address: 36:32:a2:89:68:40

Hardware Port: Thunderbolt 2
Device: en2
Ethernet Address: 36:32:a2:89:68:44

VLAN Configurations
=====
ridleylherisse@Ridleys-MacBook-Pro ~ % ifconfig en0
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=6443<RXCSUM,TXCSUM,TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
    ether 3c:06:30:33:b6:6d
    inet6 fe80::1cad:84e2:16c0:d323%en0 prefixlen 64 secured scopeid 0xc
    inet 10.0.0.207 netmask 0xfffff000 broadcast 10.0.0.255
    inet6 2601:195:c600:f9c0:1c24:dd69:2f58:8758 prefixlen 64 autoconf secured
    inet6 2601:195:c600:f9c0:888d:8ea2:a02:f88b prefixlen 64 autoconf temporary
    inet6 2601:195:c600:f9c0:c49b prefixlen 64 dynamic
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect
    status: active
ridleylherisse@Ridleys-MacBook-Pro ~ % sudo ipconfig set en0 DHCP

Password:
Sorry, try again.
Password:
Sorry, try again.
Password:
ridleylherisse@Ridleys-MacBook-Pro ~ % ifconfig en0
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=6443<RXCSUM,TXCSUM,TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
    ether 3c:06:30:33:b6:6d
    inet6 fe80::1cad:84e2:16c0:d323%en0 prefixlen 64 secured scopeid 0xc
    inet6 2601:195:c600:f9c0:1c24:dd69:2f58:8758 prefixlen 64 autoconf secured
    inet6 2601:195:c600:f9c0:888d:8ea2:a02:f88b prefixlen 64 autoconf temporary
    inet6 2601:195:c600:f9c0:c49b prefixlen 64 dynamic
    inet 10.0.0.207 netmask 0xfffff000 broadcast 10.0.0.255
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect
    status: active
ridleylherisse@Ridleys-MacBook-Pro ~ % █
```

Findings

A DHCP renewal request was performed to refresh the device's network configuration and verify DHCP communication.

5. Step 004 — Verify Connectivity After DHCP Renewal

Command Used

ping 8.8.8.8

Purpose

Used to confirm that the device maintained network connectivity after DHCP configuration changes.

Evidence

```
Last login: Mon Oct 5 20:58:01 on ttys000
ridleyherisse@Ridleys-MacBook-Pro ~ % ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=114 time=37.015 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=33.350 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=40.027 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=32.200 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=30.943 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=35.182 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=28.029 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=114 time=36.353 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=114 time=36.343 ms
^C
--- 8.8.8.8 ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 28.029/36.383/40.027/3.418 ms
ridleyherisse@Ridleys-MacBook-Pro ~ %
```

Findings

Successful connectivity testing confirmed that the device was able to communicate with external network resources after DHCP verification.

6. Conclusion

The troubleshooting process reviewed DHCP behavior, IP address assignment, and network connectivity. The device successfully obtained and maintained valid network configuration settings.

7. Skills Demonstrated

- DHCP Troubleshooting
- IPv4 Addressing
- TCP/IP Fundamentals
- Network Configuration
- Connectivity Testing
- Command Line Tools
- Technical Documentation